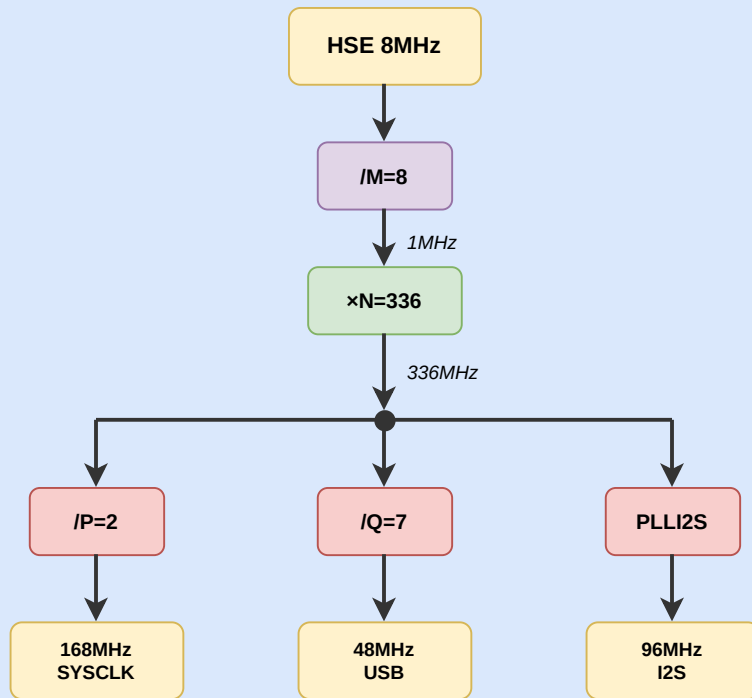


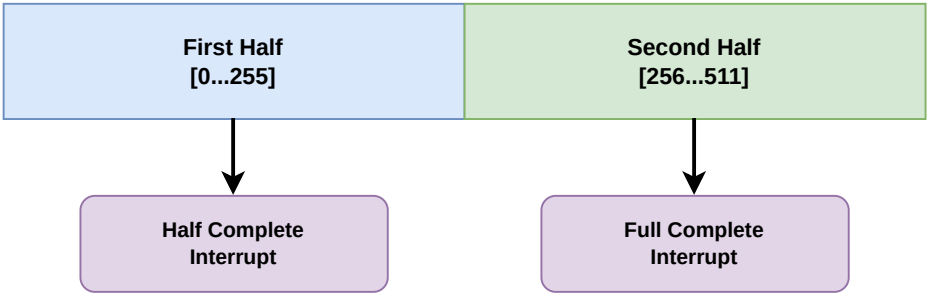
System Clock Tree



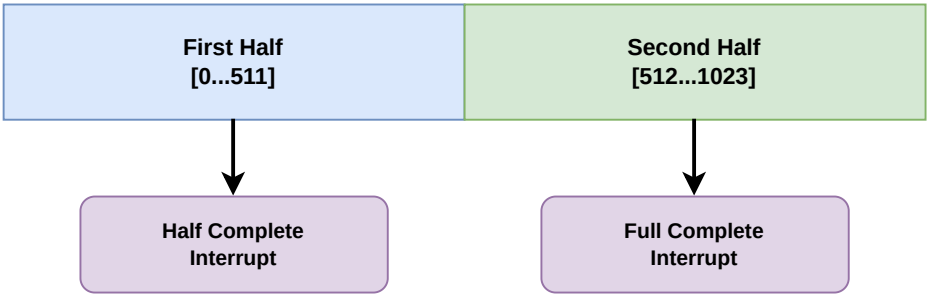
Bus Clocks	
SYSClk	168 MHz
AHB (HCLK)	168 MHz (÷1)
APB1	42 MHz (÷4)
APB1 Timers	84 MHz (×2) ← TIM2
APB2	84 MHz (÷2) ← ADC1
APB2 Timers	168 MHz (×2)

DMA Interrupt-Driven Processing

ADC Buffer (512 samples):



I2S Buffer (1024 samples - stereo):

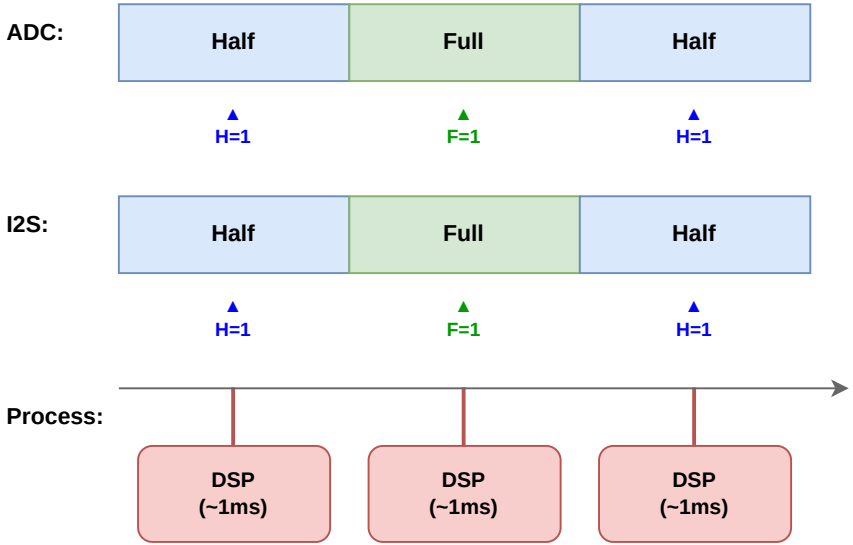


Synchronization Logic:

```
if (adc_half_ready && i2s_half_ready)
Process: adc[0-255] → i2s[0-511]

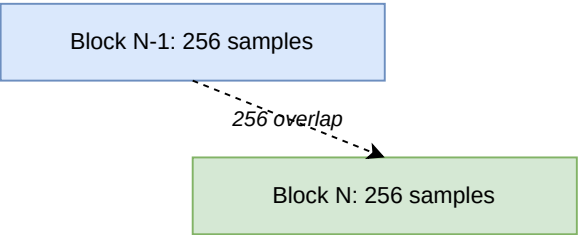
if (adc_full_ready && i2s_full_ready)
Process: adc[256-511] → i2s[512-1023]
```

Timeline:

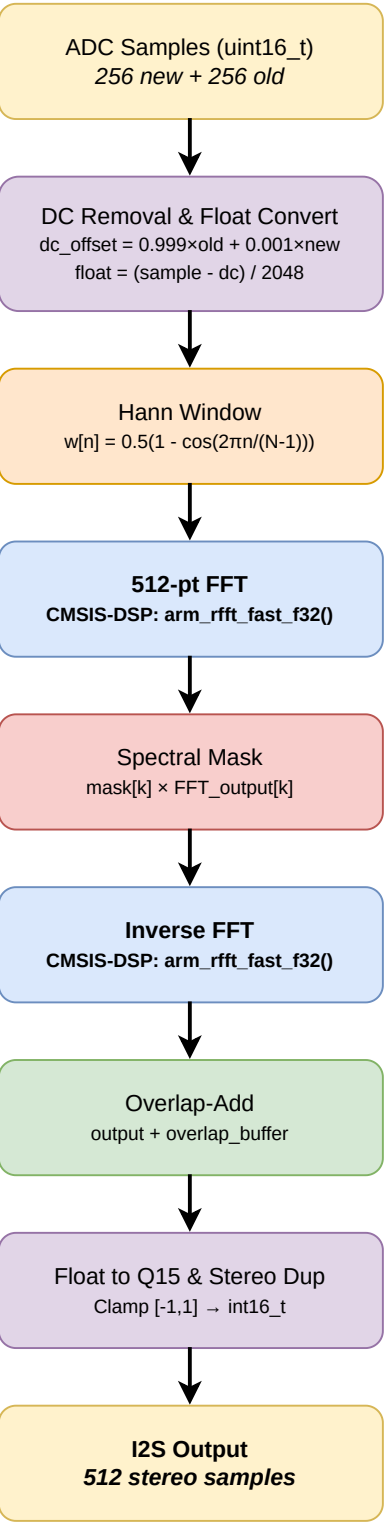


Overlap-Add FFT Processing

Time Domain Input (512 samples total)

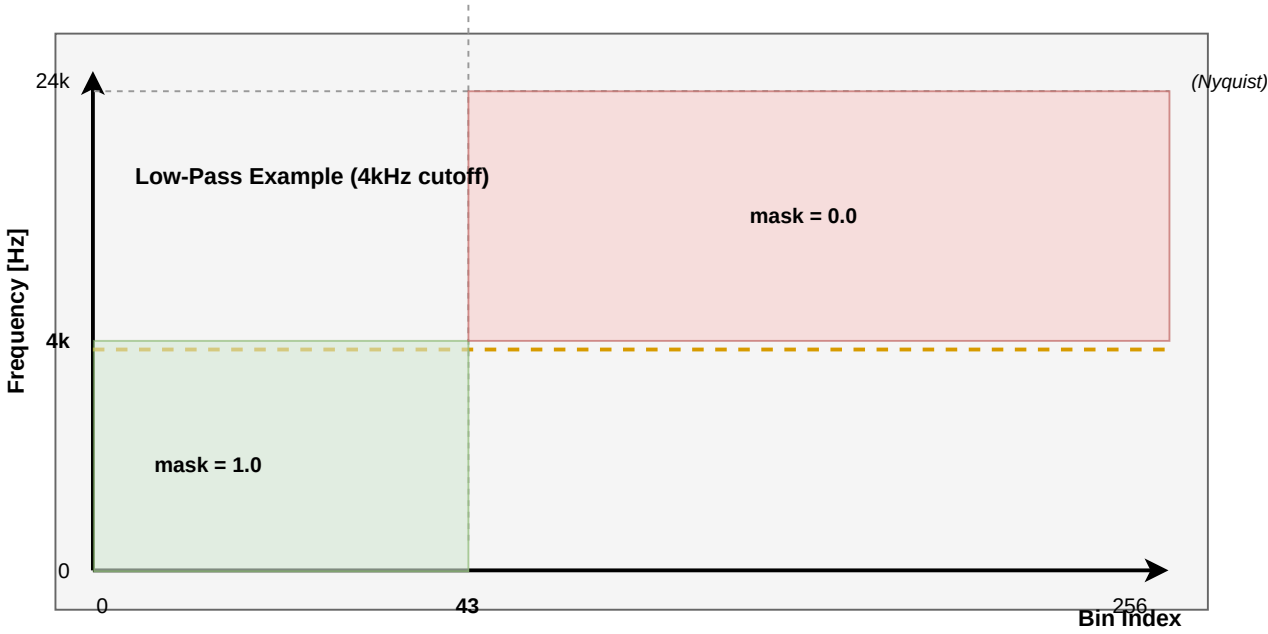


Processing Flow:



Frequency Domain Filtering

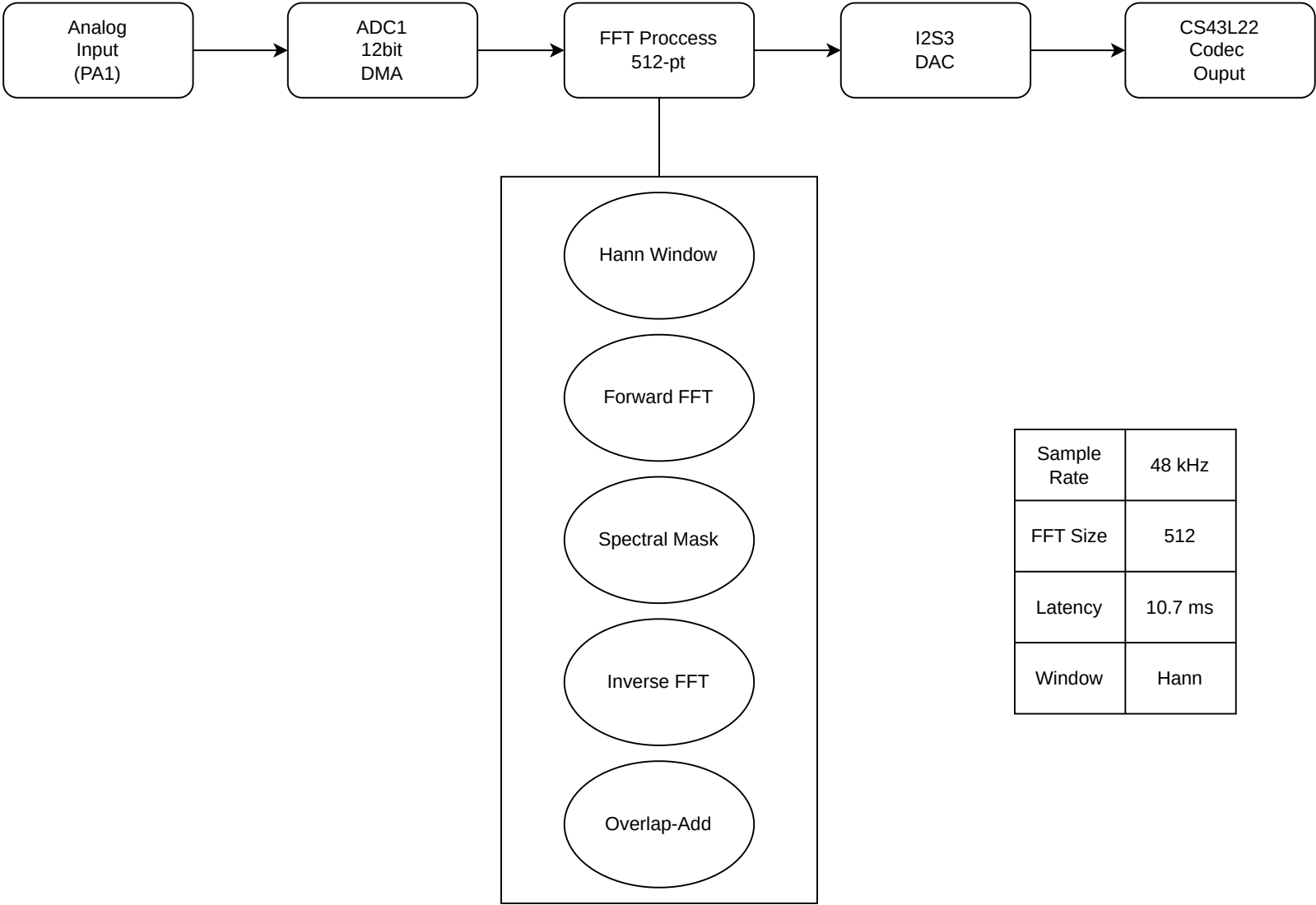
FFT Bin Resolution: 48000 Hz / 512 = 93.75 Hz/bin



Filter Examples	
Filter Type	Implementation
Low-Pass 4kHz	$\text{cutoff_bin} = \text{freq} \times 512 / 48\text{k}$ Zero bins 43-256
Band-Pass 300-3400 Hz	Keep bins 3-36 (Voice frequencies)
Notch 1kHz hum	Zero bins ± 2 around bin 11 ($1000 \times 512 / 48\text{k}$)

Code Snippet: Low-Pass 4kHz

```
// Low-pass 4kHz
float cutoff = 4000.0f;
int bin = cutoff * 512 / 48000;
for(int k=bin; k<=256; k++)
    spectral_mask[k] = 0.0f;
```



Sample Rate	48 kHz
FFT Size	512
Latency	10.7 ms
Window	Hann

Memory Buffers (All in RAM)

Buffer Name	Size	Type
adc_buffer	512	uint16_t
i2s_buffer	1024	int16_t
fft_input	512	float32
fft_output	512	float32
overlap_buffer	256	float32
input_history	256	float32
window	512	float32
spectral_mask	257	float32

Total RAM: ~14 KB for buffers

CMSIS-DSP Functions Used

arm_rfft_fast_init_f32()

Initialize 512-point Real FFT

arm_rfft_fast_f32(input, output, 0)

Forward FFT (0 = forward)

arm_rfft_fast_f32(input, output, 1)

Inverse FFT (1 = inverse)

arm_mult_f32(src1, src2, dst, len)

Element-wise multiply (windowing)

arm_add_f32(src1, src2, dst, len)

Overlap-add operation

arm_scale_f32(src, scale, dst, len)

Apply output gain

Optimization: Hardware FPU enabled

Compiler flags: -mfpv4-sp-d16 -mfloat-abi=hard

Peripheral Settings

ADC1 Configuration	
Resolution	12-bit
Channel	ADC1_IN1 (PA1)
Trigger Source	TIM2 TRGO
Sampling Time	15 cycles
DMA Mode	Circular
DMA Buffer	512 samples (uint16_t)

TIM2 Configuration (ADC Trigger)	
Clock Source	APB1 Timer (84 MHz)
Prescaler	0
Period	1749
Output Frequency	$84\text{MHz}/1750 = 48\text{ kHz}$
Trigger Output	Update Event (TRGO)

I2S3 Configuration (Audio Output)	
Mode	Master Transmit
Standard	Philips I2S
Data Format	16-bit
Audio Freq	48 kHz
MCLK Output	Enabled
DMA Mode	Circular
DMA Buffer	512 samples (int16_t x2)

Pin	Function	Peripheral
PA1	ADC Input	ADC1_IN1
PD4	Codec Reset	GPIO Output
PB6	I2C Clock	I2C1_SCL
PB9	I2C Data	I2C1_SDA
PA4	I2S WS	I2S3_WS
PC7	I2S MCLK	I2S3_MCK
PC10	I2S Clock	I2S3_CK
PC12	I2S Data	I2S3_SD

